

NUMBER COOKBOOK: NUMBER UNDERSTANDING OF LANGUAGE MODELS AND HOW TO IMPROVE IT

Haotong Yang¹²³ Yi Hu¹² Shijia Kang¹² Zhouchen Lin^{1234*} Muhan Zhang^{23*}

¹ School of Intelligence Science and Technology, Peking University

² Institution for Artificial Intelligence, Peking University

³ State Key Lab of General Artificial Intelligence, Peking University

⁴ Pazhou Laboratory (Huangpu), Guangzhou, Guangdong, China

haotongyang@pku.edu.cn {huyi2002, kangshijia}@stu.pku.edu.cn
{zlin, muhan}@pku.edu.cn

ABSTRACT

Large language models (LLMs) can solve an increasing number of complex reasoning tasks while making surprising mistakes in basic numerical understanding and processing (such as $9.11 > 9.9$). The latter ability is essential for tackling complex arithmetic and mathematical problems and serves as a foundation for most reasoning tasks, but previous work paid little attention to it or only discussed several restricted tasks (like integer addition). In this paper, we comprehensively investigate the **numerical understanding and processing ability** (NUPA) of LLMs. Firstly, we introduce a benchmark covering four common numerical representations and 17 distinct numerical tasks in four major categories, resulting in 41 meaningful combinations in total. These tasks are derived from primary and secondary education curricula, encompassing nearly all everyday numerical understanding and processing scenarios, and the rules of these tasks are very simple and clear. Through the benchmark, we find that current LLMs fail frequently in many of the tasks. To study the problem, we train small models with existing and potential techniques for enhancing NUPA (such as tokenizers, positional encoding, and number formats), comprehensively evaluating their effectiveness using our testbed. We also finetune practical-scale LLMs on our proposed NUPA tasks and find that 1) naive finetuning can significantly improve NUPA on many but not all tasks, and 2) surprisingly, techniques designed to enhance NUPA prove ineffective for finetuning pretrained models. We further explore the impact of chain-of-thought techniques on NUPA. Our work provides a more detailed and comprehensive understanding of NUPA in LLMs. Our benchmark and codes are released at https://github.com/GraphPKU/number_cookbook.

1 INTRODUCTION

The mathematical and reasoning abilities of large language models (LLMs) are currently quite impressive (OpenAI, 2023; Meta, 2024a; OpenAI, 2024a; Yang et al., 2024a), capable of solving problems at the level of a high-school student or even more difficult ones like GAOKAO (a nationwide examination of high school students applying to universities in China) (Zhang et al., 2024b), Olympiad-level problems (He et al., 2024), and college mathematics (Tang et al., 2024). However, upon closer examination of the models’ outputs, we find that although the models demonstrate remarkable proficiency in problem-solving approaches, they often struggle with basic numerical understanding and processing — like a careless student who claims, “*I know how to do it, but I didn’t get it right.*” Some of these errors are quite surprising, such as believing that $9.11 > 9.9$ or making mistakes in simple addition $8/7 + 3/5$. These errors are a major cause of hallucinations when dealing with math, reasoning, and data analysis tasks, as the model presents seemingly correct problem-solving approaches, but ultimately produces incorrect results (Huang et al., 2024; Li et al., 2024b; Jiang et al., 2024). Therefore, investigating and improving the fundamental “**numerical understanding and processing abilities**” (NUPA) of models is crucial.

*Corresponding Authors

Table 1: Task overview of NUPA Test. The four rows represent four numerical representations, and the 17 columns correspond to different tasks. ✓: 41 tasks included in our test. ✗: Not included, too complex. ○: Not directly included but can be easily adapted from an included task. —: Not applicable. The detailed explanation for these non-included tasks is provided in Appendix A.1.5

	Elementary arithmetic				Comparison			Digit Understanding						Conversion			
	Add	Sub	Multiply	Truediv	Floordiv	Mod	Max	Min	Digit Max	Digit Min	Digit Add	Get Digit	Length	Count	To Float	To Scientific	Sig. Fig.
Integer	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	—	✓	✓
Float	✓	✓	✓	✗	—	—	✓	✓	✓	✓	✓	✓	✓	○	—	✓	✓
Fraction	✓	✓	✓	✓	—	—	✓	✓	—	—	—	—	—	○	✓	○	○
Scientific	✓	✓	✓	✗	—	—	✓	✓	—	—	—	—	—	○	✓	—	○

However, in current research, **reasoning ability and NUPA are often tested together**, both on classic datasets such as GSM8k (Cobbe et al., 2021), MATH (Hendrycks et al., 2021b), MMLU (Hendrycks et al., 2021a), and in more challenging tests mentioned above. For example, a problem in GSM8k is: “Natalia sold clips to 48 of her friends in April, and then she sold half as many clips in May. How many clips did Natalia sell altogether in April and May?” Solving this problem requires two aspects: on the one hand, mathematical reasoning including understanding the text, extracting relevant information, formulating mathematical equations (or finding other solution methods), solving the equations or executing an algorithm, and obtaining the result; on the other hand, it also requires *understanding and processing the numbers* provided in the problem or produced as intermediate results at each step, like $48/2 = 24$ and $48 + 24 = 72$. While these two abilities are both essential to correctly solving the problems, tests on such datasets do not distinguish between them.

A more severe issue is that the numerical content is often **deliberately simplified** in these datasets. In various exam questions (like in the American Invitational Mathematics Examination (Li et al., 2024a)), to focus on assessing students’ understanding of mathematical concepts — such as how to set up the correct equations and apply the right theorems — the numbers in both the questions and answers are often specially chosen to be **integers**. However, this is **not** the case in **real-world scenarios** (Chen et al., 2021).

Despite the importance of NUPA, there is still a lack of accurate, detailed, and comprehensive formalization, measurement, and analysis of this fundamental capability. In this paper, we take the preliminary step towards formalizing the NUPA of LLMs. We categorize the numerical concepts and operations from primary and secondary education into four representations: *integers*, *floating-point numbers* (finite decimals), *fractions*, and *scientific notation*, along with four ability categories comprising 17 tasks. Pairing these representations results in 41 meaningful tasks, forming our NUPA benchmark (Table 1). These representations and tasks cover the most common scenarios involving number understanding and processing, which are typically not challenging for humans, as we read, use, or process such numbers nearly every day.

On this benchmark, we rigorously test several state-of-the-art LLMs containing GPT-4o (OpenAI, 2024a), Llama-3.1 (Meta, 2024a) and Qwen2 (Qwen Team, 2024). We ask the models to directly output the answers without calling external tools. Although the latest LLMs perform well on some of the easiest tasks, their performance declines significantly as tasks become slightly more complex (such as multiplication, modulus operations, or digit-based calculations), or as the representation of numbers extends beyond basic integers. See Figure 2 of Section 2.4. The overall unsatisfactory performance highlights a pronounced mismatch between the claimed strong mathematical reasoning abilities and the poor *practical, everyday* numerical understanding and processing abilities of today’s LLMs.

To address this issue, we explore three categories of approaches to enhance the NUPA of models. The first category of techniques aims at improving models’ NUPA during the *pretraining* stage, including alternative tokenization, specially designed positional encoding (PE) (Haviv et al., 2022; Kazemnejad et al., 2023; Zhou et al., 2024b), changing number formats (like zero-padding, index-hint (Zhou et al., 2024a) and reverse representation (Lee et al., 2024; Zhou et al., 2024b)). We evaluate and analyze them on our newly introduced benchmark, verifying their effectiveness/ineffectiveness on respective tasks/representations, which extends over previous evaluation mainly on the integer addition/multiplication tasks. Further, we summarize these techniques into three mechanisms: simplifying the reasoning process, aiding digit alignment, and providing regularization, and discuss the potential of these mechanisms to be applied across a broader range of numerical representations.

The second category of approaches aim to improve NUPA for an *already trained* model. We find that while simple direct finetuning can significantly enhance NUPA performance, applying the

aforementioned techniques (PEs, data formats and tokenizers) at this stage may have *adverse effects*. We test various settings and finetuning configurations, but none are able to achieve performance equal to or better than the original model. Our results suggest that these modifications can significantly disrupt the models’ established behavior or conflict with its pre-existing knowledge, leading to a decrease in performance.

Finally, we discuss the potential of using *chain-of-thought* (CoT) techniques (Wei et al., 2022) for numerical processing. Although CoT methods can break down complex problems into simpler sub-tasks and significantly increase the likelihood of obtaining correct answers, their drawbacks — such as consuming a large context window and requiring extended processing time — become particularly apparent in numerical tasks. We test a general CoT method known as RFFT (Hu et al., 2024), and find that for more complex tasks (such as multiplication and fraction addition), chain-of-thought methods face scalability challenges, making them difficult to be applied in practical scenarios. It is noteworthy that in this paper, we do not discuss tool use methods (Schick et al., 2023; Lu et al., 2023a) for NUPA as 1) we want to study the self-contained NUPA of LLMs, 2) calling external tools whenever encountering numbers increases the inference latency (Xu et al., 2024), and 3) we believe NUPA without tools is a necessary ability of AGI.

In summary, we propose a more comprehensive benchmark on the basic numerical understanding and processing abilities (NUPA) of LLMs, evaluate several SOTA LLMs’ performance on it, and further study three categories of approaches to improve NUPA: pretraining, finetuning and CoT. Our results reveal that the current research is insufficient to fully address the NUPA problem, despite it being a fundamental capability for solving many more complex tasks. We hope that by introducing a systematic classification and more comprehensive evaluation of NUPA, we can bring greater attention from the community to this important but overlooked fundamental capability.

2 NUPA TEST: A BENCHMARK FOR NUMBER UNDERSTANDING AND PROCESSING ABILITY

In this section, we will introduce our NUPA benchmark from the following four aspects: number representations, tasks, metrics, and result analysis of current LLMs. We will explain the rationale behind the inclusion (or exclusion) of specific representations and tasks in our benchmark, highlighting their distinctive features.

2.1 NUMBER REPRESENTATION

As discussed above, we believe that the *educational curricula* on the (Chinese) primary and secondary school levels serve as a valuable reference for determining the essential NUPAs that LLMs should master. We identify four number formats in these curricula that are both common and sufficient to cover most practical scenarios.

- **Integer:** The most common number and the foundation of other number representations.
- **Floating-Point Number (Float):** Floats, or finite decimals, are a useful subset of fractions. Calculations with floats like addition and comparison, work similarly to integers, making them common in daily life.
- **Fraction:** We consider fractions with integer numerators and denominators. In practical situations involving distribution, fractions become unavoidable, especially when the inaccuracy introduced by converting fractions to floats is unacceptable.
- **Scientific Notation:** Scientific notation is characterized by separating a number’s precise value from its order of magnitude. It is widely used in fields like physics, economics, and computer science because it efficiently handles a wide range of numbers and clearly conveys significant figures and precision. For LLMs, mastering scientific notation can significantly enhance their ability to handle practical tasks, such as interpreting financial reports or reading scientific texts.

Details of these four representations in our benchmark can be found in Appendix A.1.1. There are possible representations of numbers that are not included in these four formats, like *complex numbers*, *infinite decimal representation* (repeating and non-repeating), *radical expression* (like $\sqrt{2}$), ... These representations either occur infrequently in practical conversations (e.g., complex numbers) or present significant challenges for language models to process without the aid of external tools (e.g., radicals). For these reasons, we have opted not to include them in our benchmark at this stage.